

Validity Analysis: MILU

Target context: Hindi-Medium Competitive Exam Aspirants — North India (Central Exams)

Overall risk: **HIGH**

Deployment Context

Use case: Deployment scenario: A Hindi-speaking student in India is testing his/her preparation for competitive job-related examinations, with an AI system providing feedback on their responses. The goal is to evaluate the applicability of the benchmark and the quality of the LLM's feedback.

Domain: Educational assessment

Setting: Mobile application / enterprise software

Note: The deployment hypothesis should be tested using Hindi-language sentences from the benchmark, as I am familiar with the language and can provide evaluation in a subsequent stage.

Target population: A graduate student in North India interacting in the Hindi language. The student has completed their education primarily in Hindi, with limited exposure to English.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ✓	3	Moderate gaps	high
Input Content △	2	Significant gaps	high
Input Form ✓	4	Minor gaps	high
Output Ontology △	1	Serious concern	high
Output Content	2	Significant gaps	medium
Output Form △	1	Serious concern	high
Average	2.2		

△ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

MILU is the most relevant publicly available benchmark for Indic-language knowledge evaluation and provides genuine value for a portion of this deployment's needs — particularly for measuring MCQ-level accuracy on Indian Polity, History, Geography,

and Economics in Hindi. Its India-first design [Q11], real-exam sourcing [Q24, Q25], and 41-subject taxonomy [Q42] are well-suited to a central-exam preparation context in principle. However, two structural issues sharply limit its validity for THIS deployment: (1) Output form/ontology mismatch (Output Ontology=1, Output Form=1) — MILU evaluates only MCQ option selection, while the deployment requires open-ended Hindi explanatory rationales; this is a fundamental scope gap acknowledged by the paper itself [Q85] and confirmed by data inspection (DATASET-D1, D9). (2) Hindi partition content concerns (Input Content=2) — dataset analysis found 100% is_translated=True in the 26-item Hindi validation sample (DATASET-D13), heavy Engineering/Tech skew misaligned with UPSC/SSC syllabi (DATASET-D5/D6/D7/D16/D36), Current Affairs staleness (DATASET-D4/D25/D33), and pervasive cross-language item duplication. Input form (4) is the strongest dimension; output ontology and output form are the weakest. The benchmark can serve as a partial accuracy probe for a subset of priority subjects but cannot validate the deployment's core output.

Practical Guidance

What This Benchmark Measures

MILU measures MCQ-label accuracy on Hindi-language Indian-domain knowledge across 8 macro-domains and 41 subjects. For this deployment, it provides moderate evidence for a model's ability to produce correct verdicts on UPSC/SSC-style Polity, History, Geography, and Economics MCQs in Hindi (Input Ontology partial coverage, Input Form strong, Input Content partial). It offers no evidence on the deployment's core output: the Hindi explanatory rationale (Output Ontology and Output Form both score 1).

Construct Depth

Construct depth is shallow for this deployment. MILU probes whether a model can rank the correct option among up to four — a necessary but far-from-sufficient signal for exam-feedback quality. It does not probe pedagogical clarity, factual rationale accuracy, fluency, register appropriateness, or robustness to authentic Hindi-medium exam phrasing. Combined with cross-language duplication, possible high translation rate in Hindi, and Current Affairs staleness, even the verdict-accuracy signal is partially construct-irrelevant for 2025–26 central-exam readiness.

What Else You Need

Three supplementations are essential: (1) A custom Hindi-language rationale-quality evaluation framework with rubric covering factual accuracy, fluency, pedagogical clarity, and register — no public resource exists [WEB-13, WEB-14]. (2) A central-exam-specific subset of MILU filtered to UPSC/SSC/banking items with explicit Mathematics/Reasoning and Current Affairs categories, ideally drawing on natively authored Hindi items rather than translated content. (3) A current-affairs evaluation set refreshed annually with 2024–2026 events to address staleness.